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MIL-E-80238B
23 September 1989
SUPERSEDING
MIL-E-80238A
29 March 1985

MILITARY SPECIFICATION

ELECTRO-EROSION MACHINE, ELECTRO-CHEMICAL, HORIZONTAL RAM TYPE, CAVITY SINKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electro-chemical machine having a horizontal ram, hydraulic or mechanical feed.

1.2 Classification. This specification covers electro-erosion machines of the following sizes. The size required shall be as specified (see 6.2).

Size 1	10,000 ampere	12-inch stroke
Size 2	20,000 ampere	18-inch stroke

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3410

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2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears.

AGMA 2000.A88 - Gear Classification and Inspection Handbook.

AGMA 2001.B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-slots, Their Bolts, Nuts, Tongues, and Cutters. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders Association, 7901 Westpark Drive, McLean, Va 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

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(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt or remanufactured products are allowed under this specification.

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3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of 29 CFR 1910 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. During operation, service, transportation, or storage of the machine, it shall not emit materials hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter and a low lubrication level warning light that is in clear view of the operator's normal work station. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least a 24 hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts except for hand-fitted parts, such as ways, shall be interchangeable without requiring modification. Tooling and accessories normally supplied as standard shall also be considered replacement parts.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

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3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only when specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall conform to or exceed all provisions of AGMA 2001.B88, AGMA 2000.A88 and AGMA 390.03,for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a base, worktable, table work tank, work area enclosure, machining head, ram feed system, electrolyte system, electrolyte reservoir, electrolyte pressure pump, electrolyte filter, piping and fittings, electrolyte temperature control system, electrolyte mist collector, and electrolyte clarification unit.

3.4.1 Base. The base shall be designed to assure maximum rigidity and shall be stress relieved and normalized for dimensional stability. The base shall incorporate, either as an integral part or as a separate unit, provisions for accepting the table. Means shall be provided for leveling the machine.

3.4.2 Worktable. The worktable shall be fixed to the base. The table shall be fabricated of corrosion-resistant AISI 316 stainless steel and shall meet the requirements of table I. T-slots and key ways shall be machined into the table to meet the requirements of ANSI B5.1. The worktable shall be electrically insulated from the rest of the machine.

3.4.3 Table work tank. The table work tank shall be fabricated of AISI 316 stainless steel or of a material not adversely affected by all chemicals used as electro-chemical machining electrolytes.

3.4.4 Work area enclosure. The enclosure shall be constructed of AISI 316 stainless steel of a material not adversely affected by all chemicals used as

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electro-chemical machining electrolytes and shall completely enclose the work area and prevent escape of electrolyte and vapors. The machine shall have a power driven mechanism that will slide the enclosure horizontally on the base to expose the front, back and top portion of the work area. The enclosure shall have observation windows of optically clear material not adversely affected by electrolyte solution and of sufficient strength to withstand forces of high pressure electrolyte against the inside surfaces. Means shall be provided to prevent the electrolyte pump from being energized when the enclosure is open.

3.4.5 Machining head. The machining head shall be mounted on the base, opposite the work area enclosure, and shall be designed to meet the maximum rigidity requirements of table II. The head shall support a ram and platen which shall be driven horizontally the travel distance required in table I. The travel of the ram shall be parallel with the worktable surface within the tolerances specified in table II. Bearing support shall be on all four sides to provide rigidity of the ram in all positions. The ram shall be protected from electrolyte vapors, mist, or gases by a suitable enclosure with detectors installed to indicate the presence of moisture. The platen shall be made of AISI 316 stainless steel with T-slots conforming to ANSI B5.1.

3.4.6 Ram feed system. The ram shall be driven by a hydraulic cylinder or a gear train actuated by either an electric or hydraulic motor and a precision leadscrew. The feed system shall have all necessary controls to assure proper positioning of the electrode in relation to the work piece mounted on the table. The feed system shall feed and rapid traverse the ram and platen at the rates specified in table I. The control system shall be electrically integrated with the power supply and shall automatically feed the electrode to the work piece at the proper rate to maintain the preset conditions of the gap, metal removal rate, and surface finish. Means shall be provided to manually operate the feed system for set-up purposes and to control the movement of the ram with machining power either off or on.

3.4.7 Electrolyte system. The electrolyte system shall consist of a stainless steel or fiber glass tank, a high pressure pump, suction filters (between tank and high pressure pump), transfer pumps, clarification system, necessary pressure gages, valves, final filters, flow meters, heating and cooling unit and piping.

3.4.8 Electrolyte reservoir. The electrolyte reservoir and its components shall be of AISI 316 stainless steel or of materials not adversely affected by the electrolyte solutions. The capacity of the reservoir shall be divided into two compartments, one compartment having one-third of the total capacity for collecting dirty electrolyte from the machining area and the other compartment shall store clarified electrolyte from the clarifying unit. The reservoir shall be designed to provide an overflow of electrolyte from one section to the other. A sealed cover with hatches shall be provided on the reservoir with provisions for exhausting process gases.

3.4.9 Electrolyte pressure pump. The high pressure pump shall have a capacity conforming to table I and shall be of the type not affected by abrasive particles in the electrolyte. The pump shall be designed so that the motor and electrolyte seal can be renewed without removing the pump from the line. The pump system shall be designed to operate continuously without overheating the motor, the pump, or the electrolyte with the electrolyte flow control valve in the closed position. All metal components of the pump shall be fabricated of AISI 316 stainless steel. Seals shall be of materials not affected by electrolyte solutions.

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3.4.10 Electrolyte filter. The electrolyte system shall have suction filters of 100 mesh installed in the tank at the suction outlet. Pressure gages shall be provided to indicate the pressure drop across the filter to enable the operator to determine when element replacement or cleaning is necessary. There shall be a washable type filter capable of removing all solids larger than 60 micron. The filter shall be located in an accessible position to facilitate servicing.

3.4.11 Piping and fittings. All high pressure piping and fittings incorporated in the electrolyte system shall be AISI 316 stainless steel or pressure rated neoprene hose. Fiber glass or neoprene hose may be used on the low pressure side of the system. All plumbing components shall be rated and identified for maximum working pressures of the system and for use of electrolyte solutions at a temperature of 150° Fahrenheit maximum. All components shall be provided with fittings for ready removal without disturbing the plumbing.

3.4.12 Electrolyte temperature control system. The temperature control system shall consist of a heating and cooling unit capable of maintaining a set temperature at the cathode within ± 5° Fahrenheit over a range of 110° to 120° Fahrenheit with the machine operating continuously at any power level or electrolyte flow rate. All components of the heating and cooling unit having contact with the electrolyte shall be of a corrosion-resistant material not adversely affected by the electrolyte solution. An electrolyte temperature indicator for monitoring the electrolyte temperature shall be provided and installed in a prominent position in the operating area.

3.4.13 Electrolyte mist collector. A mist collector shall be installed on the machine to remove the electrolyte spray mist from the table enclosure and return the liquid to the electrolyte reservoir. Drain lines shall be installed to drain electrolyte from blowers to work enclosure.

3.4.14 Electrolyte clarification unit. The clarification unit shall be installed as a component part for the machine complete with filter press, pump, hoses, piping, valves, wiring and electric motor. The clarification unit shall have a pump for pumping dirty electrolytes from the reservoir to the filter press. The filter press shall be mounted at a height permitting gravity return of electrolytes to the clean well of the reservoir. The clarification unit shall be capable of continuously removing not less than 95 percent, by volume, of the metallic hydroxides. The filter press shall be provided with an automatic shutdown device indication when filter press is full and needs cleaning.

3.4.15 Hydraulic system. The hydraulic system shall conform to or exceed the requirements of JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.16 Electrical power supply system. Unless otherwise specified (see 6.2) the electrical system of the machine shall comply with the requirement of this paragraph. A silicon controlled rectifier type power supply unit shall be furnished with each machine. The power supply unit shall have a continuous output current rating of not less than 10,000 amperes for the 10,000 amp machine and 20,000 amperes for the 20,000 amp machine. The power supply unit shall operate on

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460 volts, 3 phase, 60 hertz, alternating current, and shall have a continuously adjustable output voltage range of 0 to 24 volts. The power supply unit shall have all the necessary controls and output characteristics capable of producing the metal removal rate, surface finish, and overcut specified in 3.6. The electrical system shall conform to NFPA 79.

3.4.16.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (ac) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.17 Power supply protection. Means shall be provided to prevent inadvertent damage to the electrode, workpiece and power supply unit in case of an abnormal current or voltage increase or decrease, or if a short circuit exists between the electrode and the workpiece. The power supply unit shall be designed to cut off the power at the cathode within 10 milliseconds from the time that an abnormal current change occurs. Electrical interlocks shall be provided to prevent operation of the electrolyte pump or power supply unit when the worktable enclosure is open and when electrolyte pressure is below a set minimum. Circuit breakers and thermal overload protectors shall be provided to prevent overheating of the power supply unit and all motors.

3.4.18 Operators control panel. An instrument control panel shall be provided to monitor the controls required for proper operation of the machine and shall include the following:

a. Voltmeter for indicating operating voltage of power supply.

b. Ammeter for indicating current at cathode or bus bar.

c. Indicator for indicating electrolyte pressure at the electrolyte manifold.

d. Pilot lights to indicate operation of each circuit.

e. Feed rate indicator of ram feed rate in inches per minute.

3.4.19 Power supply cabinet. The power supply cabinet shall have removable panels or doors to provide access to the interior components to facilitate trouble-shooting and servicing. Doors or panels shall have safety interlocks that disconnect input power to the power supply unit when doors are opened or panels removed. The power supply cabinet shall have adequate cooling facilities to prevent overheating and premature failure of electrical components within the unit. Filters shall be provided to filter incoming air and prevent entrance of conductive dusts.

3.4.20 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 115 volts.

3.4.21 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time for the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than

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1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I and table II for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Performance. The machine shall be capable of die sinking operations and reproducing complex shapes and cavities to a high degree of accuracy.

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances specified in table II.

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.10 Repair parts. Required repair parts shall be furnished as specified and fully described (see 6.2).

3.11 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.11.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)

3.11.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall provide the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating

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Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information.

Type of fluid (SAE or Military or Federal specification number)
Type of filter

3.12 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that of the highest quality prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless

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otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8.1 through 4.8.5 and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41 to 104 degrees Fahrenheit and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms and other accessories shall be inspected during the trial period. Failure of any component or accessory to operate satisfactorily shall be cause for rejection.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.7. The alignment tolerance test shall be performed prior to and subsequent to the performance test given in 4.8.4 through 4.8.5. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.4 Performance accuracy tests. The machine shall perform the tests in 4.8.4.1 , 4.8.4.2, and 4.8.4.3. Each test shall be conducted with a copper electrode and a workpiece of AISI 4340 steel. The dimensions and configuration of the electrode shall be at the option of the machine manufacturer.

4.8.4.1 Maximum metal removal rate. The machine shall be tested to demonstrate that it is capable of removing metal at the rate of 0.100 cubic inch per minute per 1,000 amperes per square inch.

4.8.4.2 Surface finish. The machine shall be tested to demonstrate that it is capable of producing a surface finish not rougher than 16 microinches when operating at the optimum metal removal rate or amount of overcut.

4.8.4.3 Overcut control. The machine shall be tested to demonstrate that it is capable of machining cavities and through-holes with electrolyte passages either

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in the electrode or workpiece. The machine shall control the electrode overcut with a tolerance not greater than 0.005 inch when operating at the optimum metal removal rate.

4.8.5 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.6 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The machines covered by this specification are for manufacturing a variety of parts where numerically controlled lathes are advantageous. The machine will automatically machine parts held in a chuck or between centers.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

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g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

i. Electrical power supply, if different (see 3.4.16).

j. Motors, if different (see 3.4.16.1).

k. Hour meter, if required (see 3.4.21).

l. Size and capacity, if different (see 3.5).

m. Specify physical size limitations, if required (see 3.5.1).

n. Specify and fully describe optional equipment, if required (see 3.9).

o. Specify applicable repair parts, if required (see 3.10).

p. Lubrication plate, if required (see 3.11.1).

q. Information to be provided on lubrication plate, if different (see 3.11.1).

r. Nameplate, if different (see 3.11.2).

s. Hydraulic plate, if required (see 3.11.3).

t. Hydraulic plate, if different (see 3.11.3).

u. Inspection conditions, if different (see 4.3).

v. First article inspection, if different (see 4.4).

w. Quality conformance inspection, if different (see 4.5).

x. Acceptance test, if different (see 4.6).

y. Additional performance test, if required (see 4.8.5).

z. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data , except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Formal 1423.

6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article item or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International D'Units (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

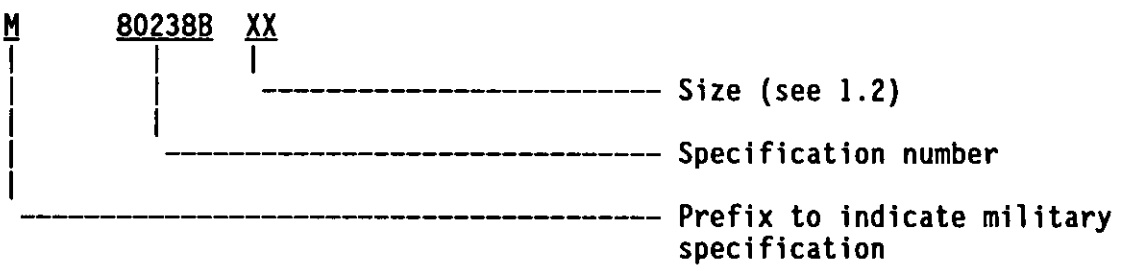
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6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



(NOTE: See MIL-STD-961C paragraph 4.4a for format.)

(NOTE: PIN must be 15 characters or less)

6.13 Subject term (keyword) listing.

Electro-chemical
Electro-erosion
Electrolyte
Electrode
Machine tool
Metal working
Die sinking

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TABLE I. Sizes and capacities.

CHARACTERISTIC	REQUIREMENTS	
	Size 1	Size 2
Ram Workhead		
Ram stroke, inches	12	18
Feed rate (infinitely variable either direction), IPM	0.020-1.0	0.020-1.0
Rapid traverse, IPM	25	25
Platen size, inches	18 x 18	18 x 18
Centerline of ram to top or table, inches	17	17
Worktable		
Table working surface, inches	42 x 42	42 x 42
Electrolyte System		
Electrolyte reservoir capacity, total gallons	600	1000
Filter press capacity, cubic feet per minute	8	16
Pump Capacity		
Gallons per minute	100	200
Pump PSI measured at outlet	275	275

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TABLE II. Alignment accuracy test.

TEST	Tolerance (not greater than)
Table level in longitudinal and traverse directions	0.0005" per foot
Table flat in longitudinal and traverse directions	0.001" per foot
Platen surface square with table surface	0.0005" per 6.000 inches
Ram travel parallel with table surface	±0.0005" per foot
Radial movement of ram during its horizontal stroke	±0.0005"
Side deflection - with the ram extended 12" and 3,000 pound force applied to the edge of the platen 90° to ram centerline, the deflection reading on the opposite side from the force shall not exceed	0.0003"
Axial deflection of ram - A 10,000 pound force shall be applied parallel to ram axis on vertical centerline and 5 inches below horizontal centerline of platen, with fully extended ram. Deflection at top and bottom of platen (indicators placed on vertical centerline 1 inch from top and bottom edges of platen) shall not exceed 0.0005 inch. The difference between top and bottom indicator reading shall not exceed	0.002"

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Custodians:
Air Force - 99

Review activities:
Army - AL
Navy - AS, MC, OS
Air Force - 84

User activities:
Army - None
Navy - AS, MC
Air Force - 84

Preparing activity:
DLA-IP

Project Number:
3410-0047

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-80238B	Electro-Erosion Machine, Electro-Chemical Horizontal Ram Type, Cavity Sinking

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: –

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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